Global Seismic Activity Pipeline

Complete Build Log

Milestone 1: Extract + Tests

TASK	WHAT YOU DID
Task 1: Project Setup + Extract	Created project skeleton, venv, and `extract.py` — fetches earthquake data from USGS API and saves raw JSON to `data/bronze/`
Task 2: Pagination + Retries	Added limit/offset pagination, exponential backoff retries on 5xx/timeout, immediate failure on 4xx
Task 3: Tests	Wrote 3 pytest integration tests — validates GeoJSON structure, bad-date error handling, and required fields on real API

Milestone 2: S3 Bronze Layer

TASK	WHAT YOU DID
Task 1: S3 Bucket + Upload	Created S3 bucket, wrote `upload_s3.py` — uploads raw JSON to `s3://bucket/bronze/<date>/raw_<date>.json` using boto3, bucket name from env var

Milestone 3: PySpark Silver Transform

TASK	WHAT YOU DID
Task 1: Bronze → Silver Transform	Wrote `bronze_to_silver.py` — PySpark reads raw GeoJSON, explodes/flattens nested fields, converts epoch-ms to timestamps, derives event_date, drops null magnitudes, writes partitioned Parquet
Task 2: Upload Silver to S3	Wrote `upload_silver_s3.py` — walks local Silver directory, uploads all Parquet files to S3 preserving partition structure, reuses `upload_to_s3`

Milestone 4: Snowflake Gold Star Schema

TASK	WHAT YOU DID
Task 1: Create Tables	Wrote `create_tables.py` — connects to Snowflake, creates dim_date, dim_location, dim_magnitude_band (pre-loaded 7 bands), fact_earthquake with surrogate keys and FKs
Task 2: Load Gold	Wrote `load_gold.py` — reads Silver Parquet, populates dims first (MERGE for idempotency), then loads facts via temp staging table + SQL JOINs for FK resolution, WHEN MATCHED UPDATE for CDC

Milestone 5: Orchestration

TASK	WHAT YOU DID
Task 1: Pipeline Runner	Wrote `pipeline.py` — one command runs all 5 steps (extract → upload bronze → transform → upload silver → load gold) with argparse for date range, status lines, timing, fail-fast error handling

Milestone 6: Reliability

TASK	WHAT YOU DID
Task 1: Multi-Day Backfill	Enhanced `pipeline.py` — splits date range into daily windows, runs each day independently, failed day logs error and continues, prints summary (succeeded/failed/total time)

Milestone 7: Deliverables

TASK	WHAT YOU DID
Task 1: README	Wrote portfolio-ready README with architecture diagram, tech stack, setup/run instructions, star schema documentation, project structure, and "What I Learned" section
Task 2: .gitignore	Created .gitignore excluding venv/, data/, __pycache__, .DS_Store
Task 3: Git + GitHub	Initialized repo, committed 16 files (778 lines), pushed to `github.com/shahzaibali1o1/earthquake-etl-pipeline`

End-to-End Flow (one command)

```
python src/pipeline.py --start 2026-01-01 --end 2026-01-05

[1/5] USGS API -> raw JSON         (extract.py - paginated, retries)
[2/5] raw JSON -> S3 Bronze        (upload_s3.py - boto3)
[3/5] raw JSON -> Parquet          (bronze_to_silver.py - PySpark)
[4/5] Parquet  -> S3 Silver        (upload_silver_s3.py - boto3)
[5/5] Parquet  -> Snowflake Gold   (load_gold.py - MERGE into star schema)
      (repeats for each day in range)
```

16 files. 778 lines of code. 7 milestones. All yours.